From Rationales to Features: Interpretable Transaction Classification via LLM Chain-of-Thought Distillation

Anonymous ACL submission

Abstract

Analyzing financial transactional data demands both predictive reliability and model transparency, yet existing approaches rarely satisfy both simultaneously. We propose a pipeline that distills Large Language Model (LLM) Chain-of-Thought reasoning into structured, human-readable features: CoT rationales are decomposed into neutral atomic claims, semantically clustered, and converted into fixed-length count vectors for classical classifiers. Across three large-scale banking benchmarks, binary gender prediction (2.6M transactions), four-class age classification (27M transactions), and churn prediction, the proposed approach consistently outperform direct few-shot LLM baselines while producing fully auditable decision logic. Unlike existing black-box techniques, every feature in our representation traces back to a human-readable behavioral statement, making the approach particularly suitable for regulated financial environments where auditability is a compliance requirement.

1 Introduction

Analysis of banking transactional data is central to risk assessment, fraud detection, marketing campaign targeting, and customer segmentation (Alves Gomes and Meisen, 2023; Mollaev et al., 2025; Padhi et al., 2021). Standard approaches rely on classical techniques such as gradient boosting, achieving strong predictive performance on expert-defined or aggregate features (Babaev et al., 2019; Pratama and Wahid, 2025; Taha and Malebary, 2020). However, these models are difficult to audit and provide limited insight into the behavioral patterns behind individual predictions (Junhai et al., 2026).

Large Language Models (LLMs) with Chain-of-Thought (CoT) prompting can generate human-readable rationales for transactional data (Shi et al., 2023; Yu et al., 2025), but remain weak as direct classifiers (Shestov et al., 2025). Raw transaction sequences are long and noisy, often exceeding context-window constraints (Abdullaeva et al., 2026; Sakhno et al., 2026a). Moreover, CoT rationales are not guaranteed to reflect the model's actual decision process: LLMs may rationalize answers post hoc or rely on implicit shortcuts (Lanham et al., 2023; Turpin et al., 2023).

We use LLM reasoning not as the final prediction mechanism, but as a source of interpretable behavioral features. Our pipeline aggregates client-level transaction statistics, generates an LLM rationale, decomposes it into neutral atomic claims, and clusters semantically similar claims into a fixed-length feature space. The resulting tabular representation can be used by classical classifiers while keeping each feature traceable to human-readable behavioral statements.

We evaluate the approach on three public banking datasets covering gender, age, and churn prediction (Boosters, 2025; Max and Valeriy, 2019; Sberbank, 2021b). Across all tasks, CoT-derived features outperform direct few-shot LLM baselines, showing that LLM rationales contain useful transactional signal. Although XGBoost models on handcrafted features or standard transaction aggregates remain strongest in raw accuracy, combining handcrafted and CoT-derived features stays close to the best classical baselines while exposing decision-relevant patterns as auditable claim clusters rather than opaque dense representations.

2 Related Work

Traditional analysis of financial transactions relies on expert-defined features and statistical aggregates fed into classical models like GBDT or logistic regression (Babaev et al., 2019). While stable for tasks such as credit scoring or churn prediction, these methods often require post-hoc tools like SHAP for limited interpretability. To improve

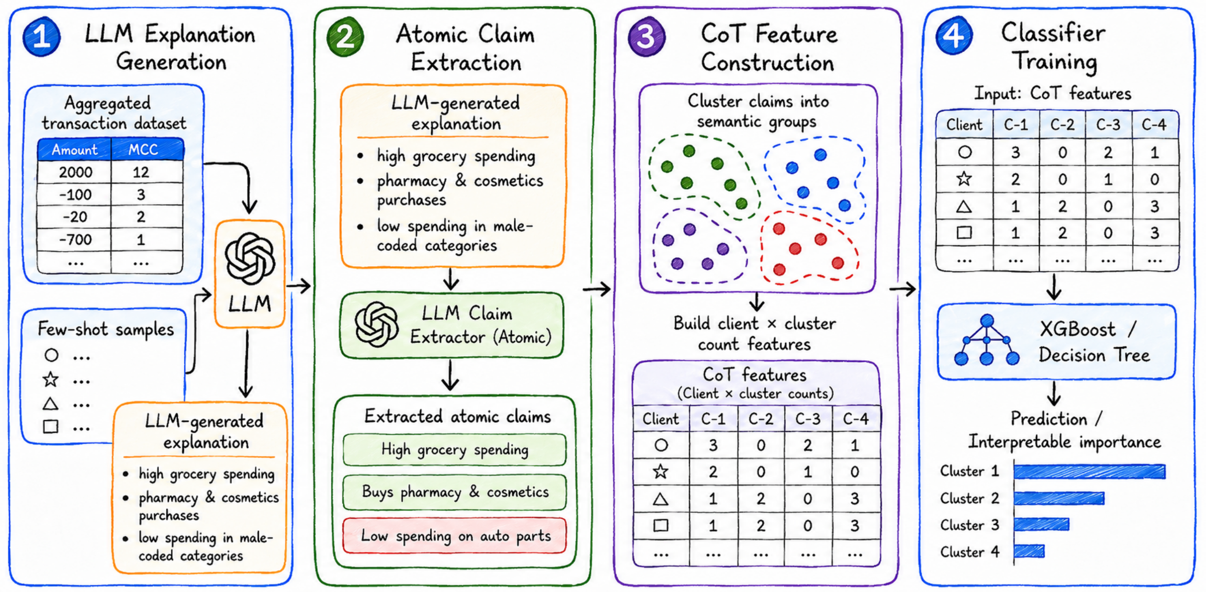

Figure 1: Overview of the proposed pipeline for extracting structured, human-readable features from LLM Chain-of-Thought reasoning. The methodology consists of four key stages: (1) LLM produces a natural language rationale based on aggregated transactional statistics; (2) Distill free-form CoT text into atomic facts; (3) Semantically similar claims are clustered to form a feature space, mapping each client's explanation to a vector of cluster frequencies; and (4) Train classical ML models on these structured features.

representations, deep learning (Padhi et al., 2021), self-supervised and contrastive learning (Babaev et al., 2022; van den Oord et al., 2018; Skalski et al., 2023) techniques extract latent embeddings from event sequences, yet these remains largely black-box in nature. Recent shifts toward LLM-based methods treat transactions as textual or event-like tokens (Shestov et al., 2025), by fine-tuning an LLM for a task with non-language data (Dinh et al., 2022) or adapting architectures like TabLLM (Hegselmann et al., 2023), or TALL-Rec (Bao et al., 2023) for abductive reasoning over behavior patterns (Shi et al., 2023; Yu et al., 2025; Sakhno et al., 2026b). However, even with CoT rationales, these systems remain opaque due to the inherent non-faithfulness and hallucination risks of generative reasoning (Lanham et al., 2023; Turpin et al., 2023). Hybrid frameworks such as LATTE (Fadeev et al., 2025), EAFD (Sakhno et al., 2026a), ESQA (Abdullaeva et al., 2026), Time-LLM (Jin et al., 2024) and FinTRACE (Sakhno et al., 2026b) attempt to bridge this gap by generating features guided by alignment and complementarity criteria with respect to pre-trained embeddings. While these methods operate in an embedding-aware loop and targets feature complementarity to existing representations, our work extends this direction by directly distilling unstructured CoT traces into a structured, numerical feature space to improve the auditability of the feature generation chain itself. Our method differs in three key respects: (1) we generate features from explicit CoT rationales rather than directly from raw sequences or embeddings, (2) our clustering operates on atomic claims extracted from those rationales, establishing a direct link between the generated features and human-readable reasoning steps, and (3) we do not rely on pre-existing embeddings as a grounding signal, making the method applicable in settings where such embeddings are unavailable or untrusted. Unlike existing solutions (Yao et al., 2018), our approach converts qualitative reasoning into human-readable features, offering a transparent alternative to latent representations without the reliability issues of raw LLM outputs.

3 Proposed Approach

Figure 1 shows our four-stage pipeline for converting LLM-generated explanations into structured, interpretable features: explanation generation, atomic-claim extraction, CoT feature construction, and classifier training.

LLM explanation generation. For each client, the LLM receives aggregated transaction statistics, dataset-level summaries, and few-shot examples, then produces a target prediction with a natural-

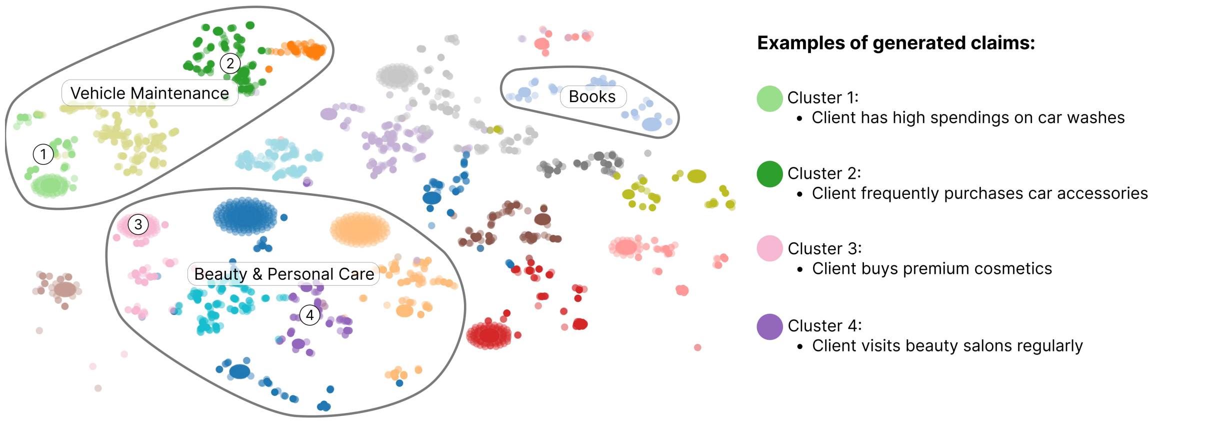

Figure 2: t-SNE visualization of atomic claim clusters from the gender dataset. Claims were clustered after PCA reduction to 50 components and t-SNE embedding with perplexity 30. Only clusters with a gender difference >0.46 were retained.

language explanation. The explanation is used to verbalize behavioral patterns observed in the transaction data rather than as the final classifier output.

Atomic-claim extraction. Each explanation is passed to an LLM-based claim-extraction step, which decomposes it into short atomic behavioral claims. Each claim describes one observable aspect of the client's financial behavior and avoids explicit target labels or multi-fact statements.

CoT feature construction. All extracted claims are embedded and clustered into semantic groups. Each cluster represents a recurring behavioral pattern. For each client, we count how many of their claims fall into each cluster, obtaining a fixed-length client-by-cluster feature vector.

Classifier training. The resulting CoT feature table is used as input to classical classifiers such as XGBoost or Decision Trees. Since each feature corresponds to a cluster of human-readable claims, the downstream model can be inspected through feature importance for XGBoost or decision rules for trees, instead of relying solely on opaque dense representations.

4 Experimental Setup

4.1 Datasets

We evaluate the proposed method on three publicly available transactional classification datasets: (1) a binary **gender classification dataset** from Kaggle (Max and Valeriy, 2019), with 2.6M transactions from 8.4K clients; (2) a four-class **age prediction dataset** from the Sberbank Sirius competition (Sberbank, 2021b), with 27M transactions from 30K clients; and (3) Rosbank Churn (Boosters, 2025), a binary **churn prediction dataset** with 0.5M transactions from 5K clients.

All datasets share a common event-sequence structure, including timestamps, transaction amounts, and categorical transaction descriptors. We split the datasets by user ID into training, validation, and test subsets using a 70/10/20 ratio. The resulting splits are approximately class-balanced: the gender dataset has a 57%/43% class split, the four age classes each account for approximately 24–26% of the data, and Rosbank Churn has a 55.3% positive rate by client.

4.2 Baselines

We compare our approach against models that represent different trade-offs between predictive accuracy and interpretability. All test results are summarized in Table 1.

LLM Baselines. We evaluate GPT-OSS-120B (OpenAI, 2025) and DeepSeek-R1 (DeepSeek-AI, 2025) using the same few-shot prompting protocol and client-level summary statistics as in Section 3.

LoRA Fine-tuning. We fine-tune Qwen models of different sizes with Low-Rank Adaptation (LoRA) (Hu et al., 2022) and a classification head. These models use the same textual input format as the LLM baselines. LoRA provides a strong black-box LLM baseline, but it returns only a final prediction and does not expose a human-readable decision structure.

Classical ML Models. We train XGBoost models on four feature sets: (1) *CoT-derived features*, generated by our claim extraction and clustering pipeline; (2) *standard transaction aggregates*, a strong non-LLM baseline built directly from trans-

Method	Gender	Age	Churn
Majority vote	0.575	0.260	0.553
Interpretable			
GPT-OSS-120B	0.628	0.338	0.521
DeepSeek-R1	0.650	0.311	0.526
XGB + CoT features	**0.683**	**0.413**	**0.605**
Non-interpretable			
Qwen-1.5B + LoRA	0.690	0.533	0.653
Qwen-7B + LoRA	0.675	0.534	0.661
Qwen-14B + LoRA	0.677	0.540	0.675
XGB + std. aggregates	0.774	**0.642**	0.743
XGB + handcrafted	**0.793**	0.600	**0.744**
XGB + handcrafted + CoT	0.786	0.601	0.739

Table 1: Test accuracy across three transactional classification datasets.

action fields; (3) *handcrafted behavioral features*, manually engineered and optimized over several iterations; and (4) a *concatenation* of handcrafted and CoT-derived features. Standard aggregates serve as a high-accuracy tabular reference point, whereas CoT-derived features prioritize auditability through their direct link to natural-language behavioral claims.

To ensure a fair comparison, we optimize all hyperparameters on the validation split over 10 independent runs. Detailed hyperparameter settings are provided in Section A.1.

5 Experimental Results

Table 1 reports test accuracy across the three datasets. The main pattern is that CoT-derived features consistently improve over direct few-shot LLM inference. Compared with the stronger few-shot LLM baseline, XGBoost on CoT features gains 3.3 points on Gender (0.683 vs. 0.650), 7.5 on Age (0.413 vs. 0.338), and 7.9 on Churn (0.605 vs. 0.526). This suggests that LLM rationales contain useful behavioral signal, but this signal is better exploited after distillation into structured tabular features.

LoRA fine-tuning is a stronger black-box LLM baseline, reaching 0.690, 0.540, and 0.675 accuracy on Gender, Age, and Churn. However, the best LoRA models still trail the strongest XGBoost baselines by 10.3, 10.2, and 6.9 points, respectively. Thus, parameter-efficient fine-tuning helps adapt LLMs to transactional classification, but does not close the gap to specialized tabular models.

The best raw accuracy is achieved by XGBoost on handcrafted behavioral features or standard transaction aggregates. Nevertheless, combining handcrafted and CoT-derived features preserves most of this performance: the combined representation is within 0.7 points of the best baseline on Gender (0.786 vs. 0.793), 4.1 on Age (0.601 vs. 0.642), and 0.5 on Churn (0.739 vs. 0.744). We therefore position the proposed pipeline as an accuracy–interpretability trade-off: it remains close to strong transaction-based ML while exposing decision-relevant patterns as human-readable claim clusters.

5.1 Effect of Cluster Filtering and Claim Quality

We further examine the CoT feature space on the Gender dataset. Figure 2 visualizes extracted atomic claims with `t-SNE`; for readability, it shows clusters with a gender difference above 46%. The clusters form coherent behavioral categories, such as "Beauty & Personal Care" and "Vehicle Maintenance", and capture concrete patterns such as cosmetics purchases or car-related spending. Dense regions contain near-duplicate claims, suggesting that repeated qualitative observations are converted into stable cluster-count features.

Figure 3 illustrates a decision tree trained on CoT-derived features. Since each split corresponds to a cluster of atomic behavioral claims, the resulting rules can be inspected by a human analyst. This does not make the classifier fully transparent in a formal sense, but shows that the distilled feature space supports more auditable downstream models than latent embeddings or end-to-end fine-tuned LLM classifiers.

6 Conclusion

We presented a pipeline that transforms LLM-generated rationales into structured, interpretable features for financial transaction classification. By distilling explanations into clustered atomic claims, our method consistently improves over direct few-shot LLM inference across gender, age, and churn prediction tasks. The resulting features remain human-readable and can be used with standard classifiers such as XGBoost, exposing decision-relevant behavioral patterns rather than opaque dense representations. While strong aggregate and handcrafted baselines still achieve the highest raw accuracy, our approach offers a competitive and auditable feature-generation layer for financial modeling.

Limitations

The proposed methodology and experimental setup have several limitations.

The evaluation focuses on three transactional prediction tasks from the banking domain: gender, age, and churn prediction. Although these tasks cover both demographic and behavioral targets, they do not establish transfer to other domains or to other financial objectives such as credit scoring, fraud detection, or risk assessment.

The extraction of atomic facts depends heavily on the quality and consistency of LLM-generated explanations. If the CoT contains hallucinations or inconsistencies, errors can cascade into atomic claims and clustering, reducing interpretability. We do not yet incorporate CoT correction or consistency checks (e.g., self-consistency filtering or post-hoc validation), which could mitigate this sensitivity.

The clustering process relies on semantic similarity metrics and fixed thresholds, which may merge unrelated facts.

Finally, while the approach measures the alignment between explanations and true labels, it does not yet evaluate their faithfulness to the actual decision process of the underlying classifier.

Future work should focus on testing the developed methodology on other domains and tasks, as well as decreasing the number of hyperparameters required to run the current version of the pipeline.

Ethics

Our experiments use only publicly available, open-source datasets released by their original providers, without additional post-processing or re-identification beyond the published splits. We acknowledge that gender and age prediction from financial transactions raises fairness concerns: such predictions could enable discriminatory profiling if deployed in production. We use these tasks solely as established interpretability benchmarks where ground-truth labels are available for quantitative evaluation, following prior work (Shestov et al., 2025; Sakhno et al., 2026a). Our pipeline is task-agnostic and equally applicable to fairness-neutral targets such as churn prediction. We do not endorse or encourage gender or age inference in real-world financial systems. All datasets were used in compliance with their published terms.

Computational experiments. For computational experiments we used NVIDIA A100 GPU with 80 Gb VRAM. Each experiment can be reproduced on a single A100 GPU in under 24 hours. We used open-source LLMs.

Use of AI Assistants. We utilize Grammarly and some LLMs (ChatGPT, Deepseek V4) to enhance and proofread the text of this paper, correcting grammatical, spelling, and stylistic errors, as well as rephrasing sentences and preparing Figure 1. Consequently, certain sections of our publication may be identified as AI-generated, AI-edited, or a combination of human and AI contributions.

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A Appendix

This appendix collects supporting figures, hyperparameters, and prompts that supplement the main text. We group materials by purpose to make navigation and cross-referencing easier.

A.1 Hyperparameters

LLM baselines were run locally using VLLM with the following settings ($temperature = 0.8$, $top\_p = 0.9$). For classical models, XGBoost is trained on handcrafted features, CoT-derived cluster features, and their concatenation.

To convert free-form CoT into structured features, we tune the clustering and classifier hyperparameters over fixed ranges. Specifically, we vary $n\_clusters \in [50, 500]$, $n\_estimators \in [10, 300]$, $max\_iter \in [50, 500]$, and use $learning\_rate = 0.1$.

Hyperparameter	Value
LoRA rank (r)	8
LoRA scaling factor (α)	8
LoRA dropout	0.1
Learning rate	2×10^{-4}
Batch size	32
Number of epochs	5

Table 2: LoRA hyperparameters used for fine-tuning

A.2 Supplementary Figures

This subsection provides visual evidence supporting the main claims and highlights how design choices affect the extracted features.

Figure 3 illustrates how a small set of CoT-derived features yields a compact, human-readable decision structure.

Figure 4 shows the sensitivity of the retained claim set to the filtering threshold.

A.3 Interpretability vs. Quality Tradeoff

We compare baselines not only by predictive quality but also by how directly their decision processes can be inspected: Majority Vote is fully transparent but only reflects the marginal label distribution; few-shot LLM classifiers provide natural-language rationales that are not guaranteed to be faithful; decision trees are inherently interpretable via explicit decision rules; and XGBoost typically requires post-hoc feature attribution. Our CoT-derived feature representation aims to narrow this gap by translating free-form model reasoning into a structured, human-readable feature space.

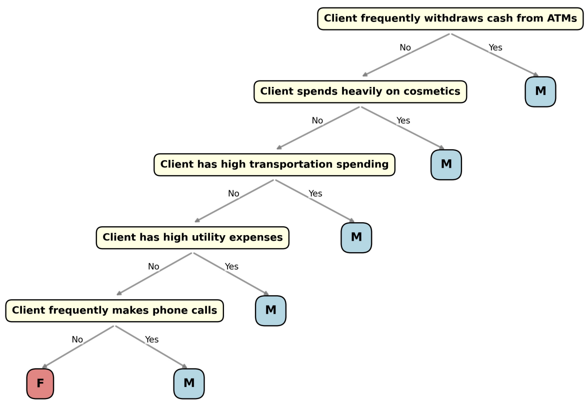

Figure 3: Decision tree fitted on CoT features produced by our pipeline. Hyperparams: $max\_depth : 7, ccp\_alpha : 0.0055$.

Figure 5 summarizes the relative efficiency of each baseline under the interpretability–quality tradeoff, situating our approach among deterministic alternatives.

A.4 Prompt Templates

For completeness, we include the exact prompt templates used in the pipeline.

A.4.1 Gender Dataset Prompt Templates

All prompt templates in this subsection correspond to the gender classification dataset.

Effect of Cluster Filtering and Claim Quality. We further analyze the internal structure of the CoT feature space on the Gender dataset. Not all extracted claim clusters are equally informative:

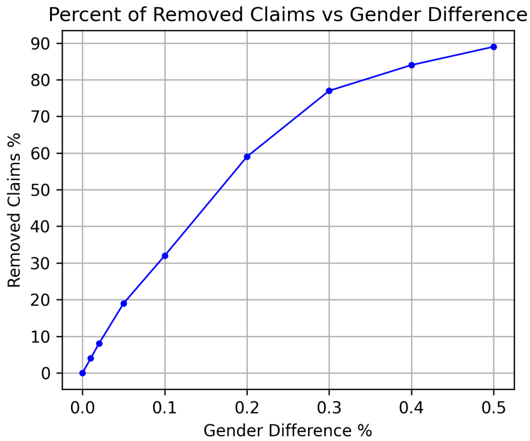

Figure 4: Percentage of removed claims from the training set as a function of the gender difference threshold used for filtering.

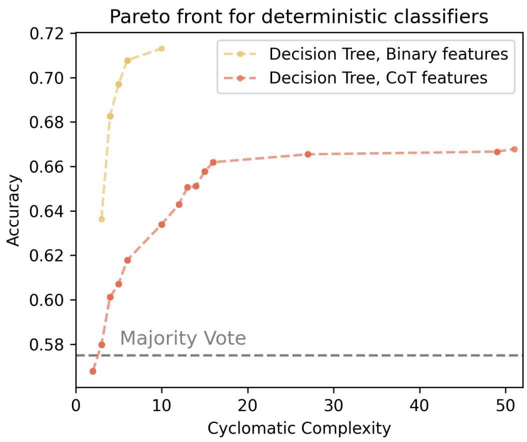

Figure 5: Pareto front illustrating the tradeoff between predictive quality and interpretability across deterministic baselines.

some mix claims from different classes and contribute little to downstream classification. Filtering clusters by class-proportion difference, as shown in Figure 12, slightly improves performance, indicating that cluster purity affects feature quality.

System prompt

```
You are an expert in analyzing customer behavior through their financial transactions.
Your goal is to reason and explain why a client belongs to a particular gender based on
their transactions and aggregated data from all users.

- Translate statistics into behavioral patterns - regularity, periodicity, activity
  spikes, etc.
- All conclusions must be based on comparing the user's behavioral patterns with
  aggregated information from the entire dataset
- It is forbidden to make conclusions that do not directly follow from data analysis
- Note key features and the degree of stability of financial behavior
- Formulate thoughts clearly and concisely
```

User prompt

```
1. Summary data for the entire dataset with user transactions, divided into male and
   female groups:

{SUMMARY_TRANSACTIONAL_STATS}

2. User examples: {FEW_SHOT_EXAMPLES}
3. Aggregated target client data: {USER_STATS}

Task:
- Identify key features of the user's behavior
- Provide a detailed explanation of what the client's true gender is, based on
  comparing the client's behavioral patterns with the average behavior of clients of
  each gender
- Do not make guesses, all conclusions must be logical and justified
- At the end of the reasoning, be sure to indicate the gender in the format:
  Final: \boxed{Male}  or  Final: \boxed{Female}.
```

Figure 6: Explanation generation prompt (system and user parts) for the gender classification dataset (Max and Valeriy, 2019).

```
Dataset summary stats:

# Men

## Summary statistics of male transaction volumes:

* Normal distribution of transaction counts: mean=482, std=1789
* Normal distribution of median transaction: mean=-21541 (expenses), std=83285; mean=41659 (income),
* Normal distribution of total expenses: mean=-16038092, std=47549744
* Normal distribution of total income: mean=4991453, std=27671505
* Normal distribution of minimum transaction: mean=-1752486, std=5526480 (expenses)
* Normal distribution of maximum transaction: mean=-3060, std=8668 (expenses); mean=1942321, std=608

## Average number of transactions by MCC codes (men, 90th percentile)
| MCC (men) | Average number of transactions |
|-----------|--------------------------------|
| Financial institutions — automatic cash withdrawal | 104 |
| Financial institutions — manual cash withdrawal | 80 |
| Grocery stores, supermarkets | 77 |
| Telephone calls | 67 |
| Money transfers | 55 |
| Gambling transactions | 42 |
| Various food stores | 37 |
| Service stations | 25 |
| Restaurants, diners | 17 |
| Public eateries, restaurants | 17 |
| Limousines and taxis | 17 |
| Department stores | 15 |
| Securities: brokers/dealers | 15 |
| Record stores | 14 |
| Galleries/video games | 13 |
| Alcohol stores (takeaway) | 13 |
| Direct marketing — catalog sales | 13 |
| Road and bridge tolls | 12 |
| Direct marketing — inbound telemarketing | 11 |

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# Women

## Summary statistics of female transaction volumes:
<...>

## Average number of transactions by MCC codes (women, 90th percentile)
<...>
```

Figure 7: Dataset-level summary statistics (SUMMARY_TRANSACTIONAL_STATS) for the gender classification dataset (Max and Valeriy, 2019), used in explanation-generation prompts.

```
# Example: Aggregated statistics for a single client

* Active period: 328.0 days
* Total transactions: 657
* Average transactions per day: 2.003

* Distribution of transactions by MCC codes:
Financial institutions — automatic cash withdrawal = 239
Grocery stores, supermarkets = 115
Telephone calls using magnetic stripe readers = 59
Cosmetic stores = 37
Various food stores — markets, specialty stores, ready-to-eat meals, sales via vending machines = 26
Pharmacies = 20
Public eateries, restaurants = 14

* Distribution of spending by MCC codes:
Various specialized household goods stores = -1123
Lumber and building materials = -1348
Telephone calls using magnetic stripe readers = -2177
Fabrics, upholstery, curtains, blinds = -3268
Public eateries, restaurants = -6691
Restaurants, diners = -7215
Home goods = -7562
Grocery stores, supermarkets = -9530
Cosmetic stores = -9742
Pharmacies = -12022
Various food stores — markets, specialty stores, ready-to-eat meals, sales via vending machines = -12
Service stations = -14597
Sporting goods stores = -14598
<...>

* Total income: 17077854
* Average income per transaction: 218947

* Total expenses: -29633111
* Average expense per transaction: 218947
```

Figure 8: Client-level aggregated transaction statistics example for the gender classification dataset (Max and Valeriy, 2019).

System prompt

```
You are provided with a Chain-of-Thought  containing model reasoning about client
behavior based on their transactions. Your task is to extract atomic behavioral
properties from the reasoning chain that reflect the characteristics of the client's
transactions. These properties are semantic features suitable for subsequent clustering.

Each atomic fact should express a qualitative characteristic of behavior, without
using numerical indicators or explicit reference to the client's gender and pronouns
he/she.

Strictly prohibited:
* Using any numbers (including percentages, amounts, quantities, ranges, ranks,
  precise time intervals, dates, seasons, holidays, months, days of the week).
* Mentioning gender and pronouns: no fact should contain information about the
  client's gender or use pronouns; facts should be expressed neutrally and
  unambiguously.
* Combining multiple properties in one fact.
* Using words: combination, simultaneously, along with, together with, and at the
  same time, and also (e.g., forbidden to generate facts like: The client almost
  never spends on pharmacies, cosmetics, and children's goods.)
* Describing absence of behavior, absence of expenses, or absence of activity.
Cannot use constructions like: client does not do, client does not spend, client
does not use, client does not perform, client does not exhibit, client avoids,
client is absent in the category, client does not have, no transactions, and any
formulations with the particle not that express absence of action or property.

Each atomic fact must follow these rules:
* One fact per element: each fact contains only one specific statement or property.
* Only one expense/income category or one direction of activity is allowed per fact.
* Each fact should describe positive presence of a behavioral trait, not its absence.
* Subject-predicate: each fact must have a clear connection between subject and
  predicate. The subject is always the client whose transaction data is provided.
* Focus on financial behavior: facts must be directly related to the client's
  financial behavior. Cannot include information unrelated to transaction statistics.
* Comprehensibility without context: each fact should be self-sufficient and
  understandable without additional information.
* All facts must be written in English.

As a result, you need to obtain a list of atomic facts that comply with the
specified rules.
The response must be in the format of a JSON list of atomic facts:

Atomic facts:
[ "fact 1", "fact 2", ... ]
```

Figure 9: System prompt for atomic-claim extraction in the gender classification setting (Max and Valeriy, 2019), specifying rules and constraints for behavioral facts.

User prompt

```
Below is a Chain-of-Thought  model reasoning about a client's transactions.
Extract atomic facts from it strictly according to the instructions.
Return the answer in the format of a JSON list of strings in English.

Example 1:
CoT: "The client performed 132 operations over 113 days, of which 74 were cash
withdrawals at ATMs. At the same time, the client did not make purchases in
supermarkets, pharmacies, or restaurants."
Atomic facts:
[
  "The client is focused on cash withdrawals.",
  "The client practically does not spend money on everyday purchases.",
  "The client exhibits narrow specialization in operation types."
]

Example 2:
CoT: "The client conducted 855 operations over 335 days. Among them, small regular
purchases predominated, as well as several large one-time expenses, including
airline tickets and tire purchases."
Atomic facts:
[
  "The client exhibits high activity in everyday expenses.",
  "The client makes small regular purchases.",
  "The client spends large amounts on airline tickets."
]

Break down the following CoT text into a list of atomic facts about the client,
following the rules specified in the system message.
CoT: {}
```

Figure 10: User prompt for atomic-claim extraction in the gender classification setting (Max and Valeriy, 2019), with examples and task specification.

System prompt

```
You are a summarizer of atomic behavioral facts.
Your task is to take a list of atomic facts about users and return one short fact
that summarizes the most common patterns while ignoring rare outliers.
The output must be a single concise sentence and must not mention gender.
```

User prompt

```
Summarize the following group of facts: {FACT_LIST}

Instructions:
- Output a single concise fact derived from the most frequent patterns.
- Ignore rare or unique facts.
- Do not mention gender.
- Return only the final summarized fact with no additional commentary.
```

Figure 11: Cluster-summary prompt for the gender classification setting (Max and Valeriy, 2019), producing one representative fact per cluster.

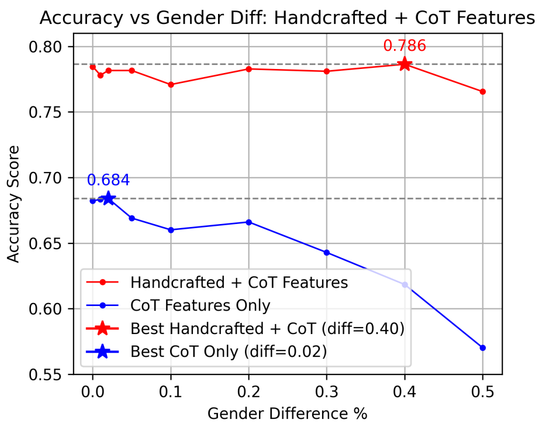

Figure 12: Accuracy vs Gender Difference Threshold when using Handcrafted + CoT Features (red) compared to CoT Features Only (blue).